

EE214 Lab-2 Report

Name: Medhansh

Roll No.: 25B3986

Date: 14/8/25

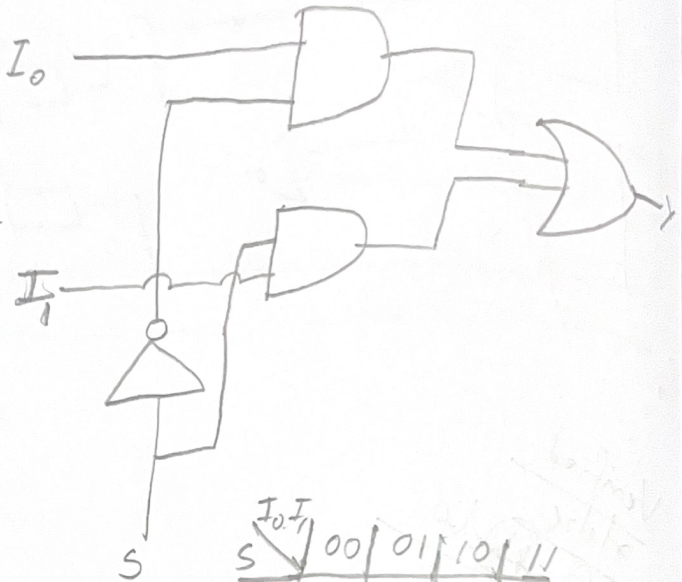
Name: Medhansh
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Part - A

Q1)

b) $Y = I_0 \bar{S} + I_1 S$

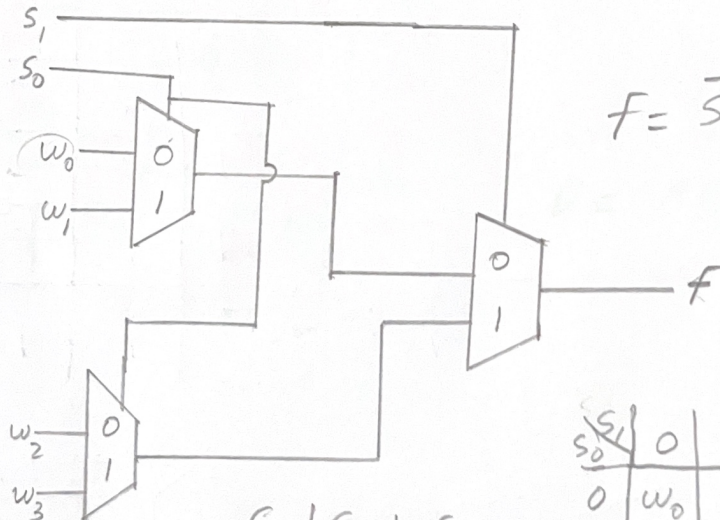
S	I_0	I_1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



$I_0 I_1$	00	01	10	11
S	0	0	1	1
0	0	0	1	1
1	0	1	0	1

Q2) Part - B

a)



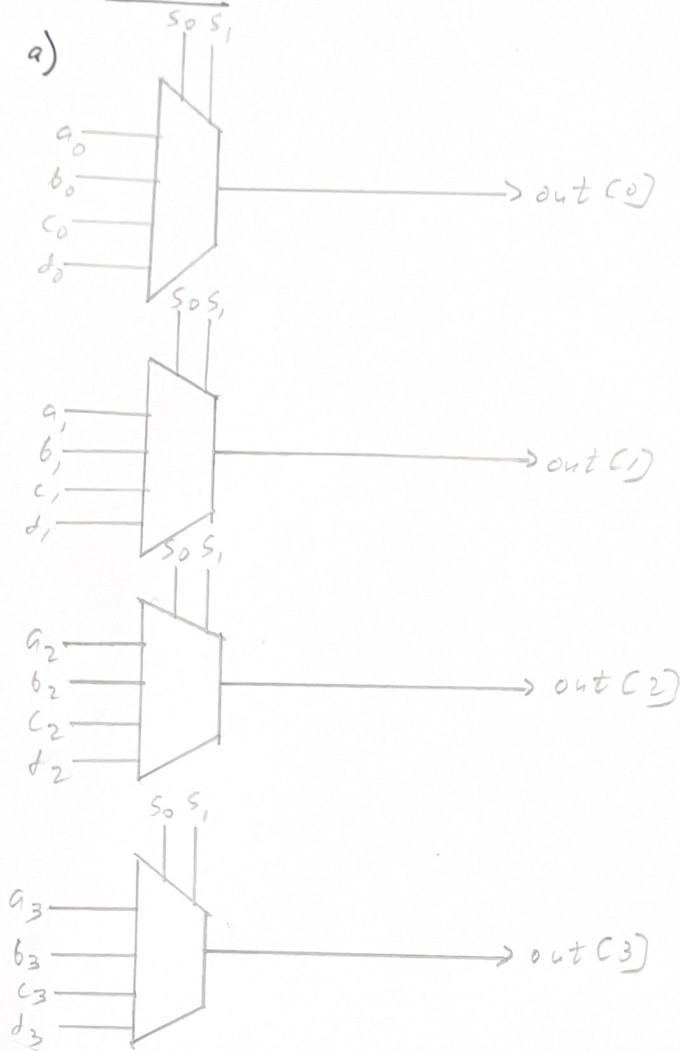
$$F = \bar{S}_0 \bar{S}_1 w_0 + S_0 \bar{S}_1 w_1 + \bar{S}_0 S_1 w_2 + S_0 S_1 w_3$$

S_0	S_1	f
0	0	w_0
1	0	w_1
0	1	w_2
1	1	w_3

S_1	0	1
S_0	w_0	w_2
1	w_1	w_3

Q3) Part-C

a)



s_1	s_0	a	b	c	d	Y
0	0	1	x	x	x	a
0	1	x	1	x	x	b
1	0	x	x	1	x	c
1	1	x	x	x	1	d

$s_0 \backslash s_1$	0	1
0	a	c
1	b	d

$$Y = \bar{s}_0 \bar{s}_1 a + s_0 \bar{s}_1 b + \bar{s}_0 s_1 c + s_0 s_1 d$$

Verified
Dev Help
14/8/26

Part A: 2x1 MUX

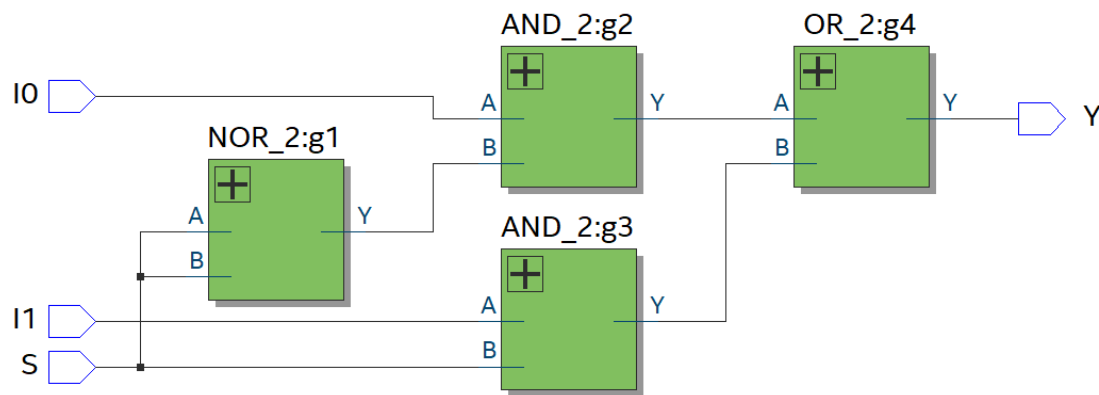


Figure 1: 2x1 Multiplexer schematic

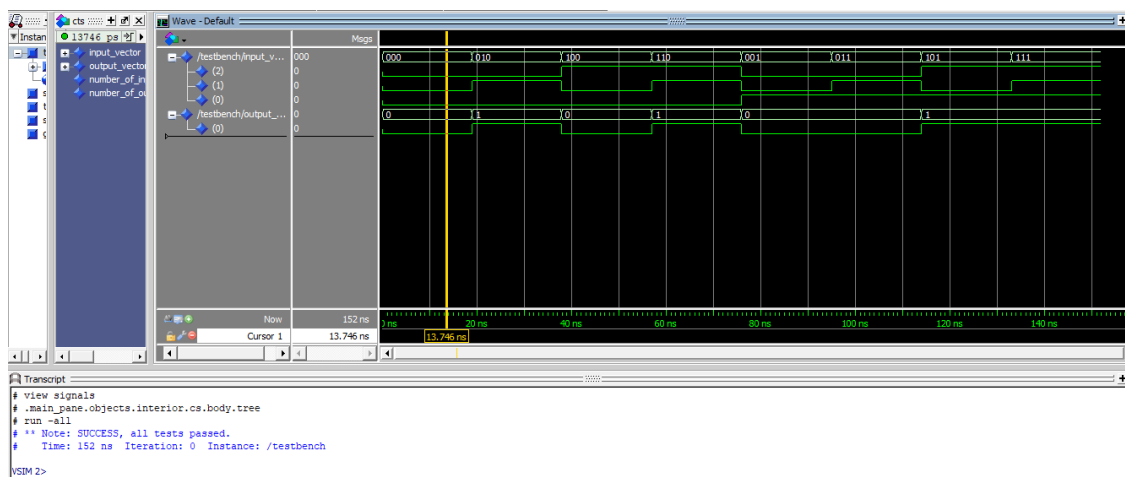


Figure 2: 2x1 Multiplexer waveform

Part B: 4x1 MUX

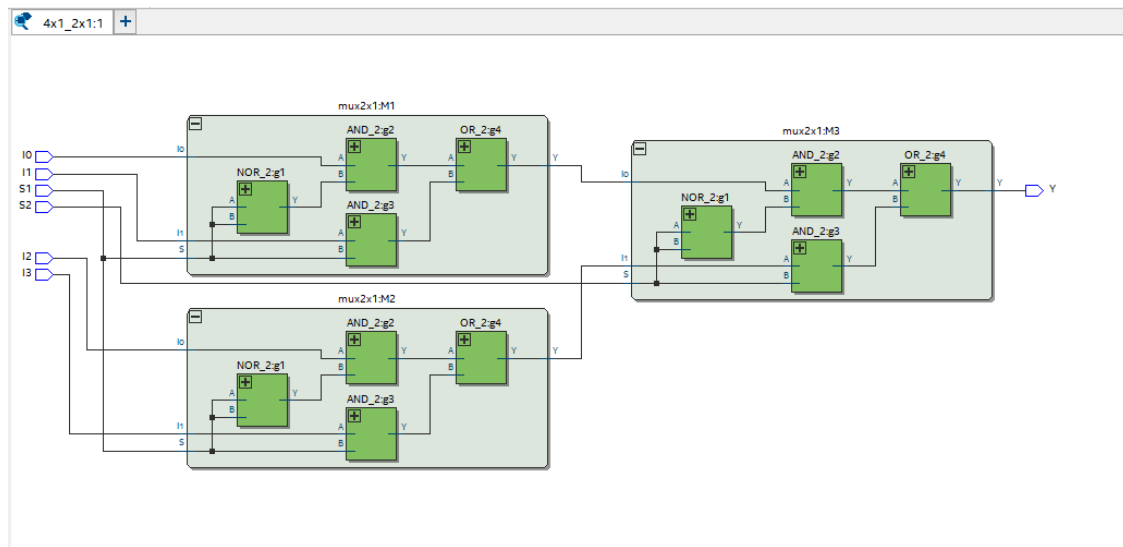


Figure 3: 4x1 Multiplexer using 2x1 Multiplexers schematic

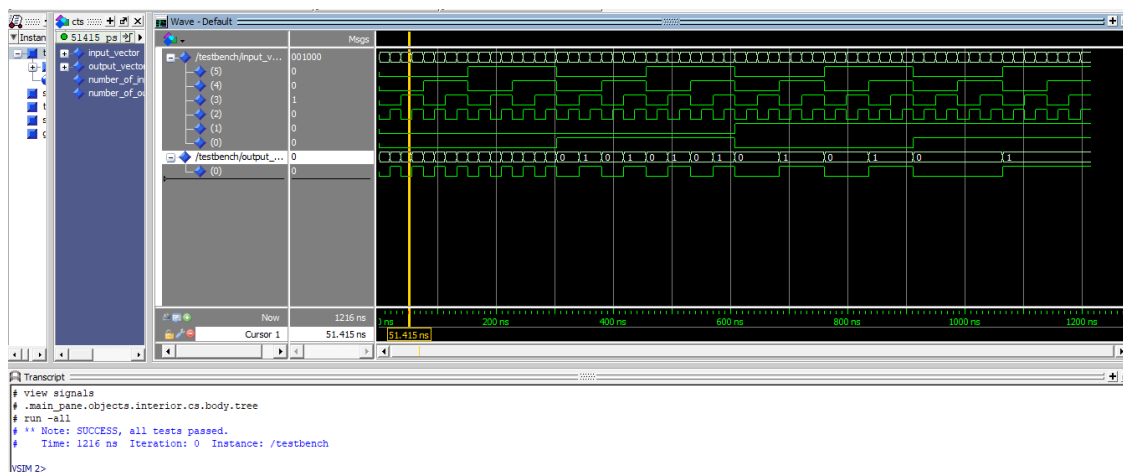


Figure 4: Corresponding waveform

Part C: 4-bit 4x1 Multiplexer

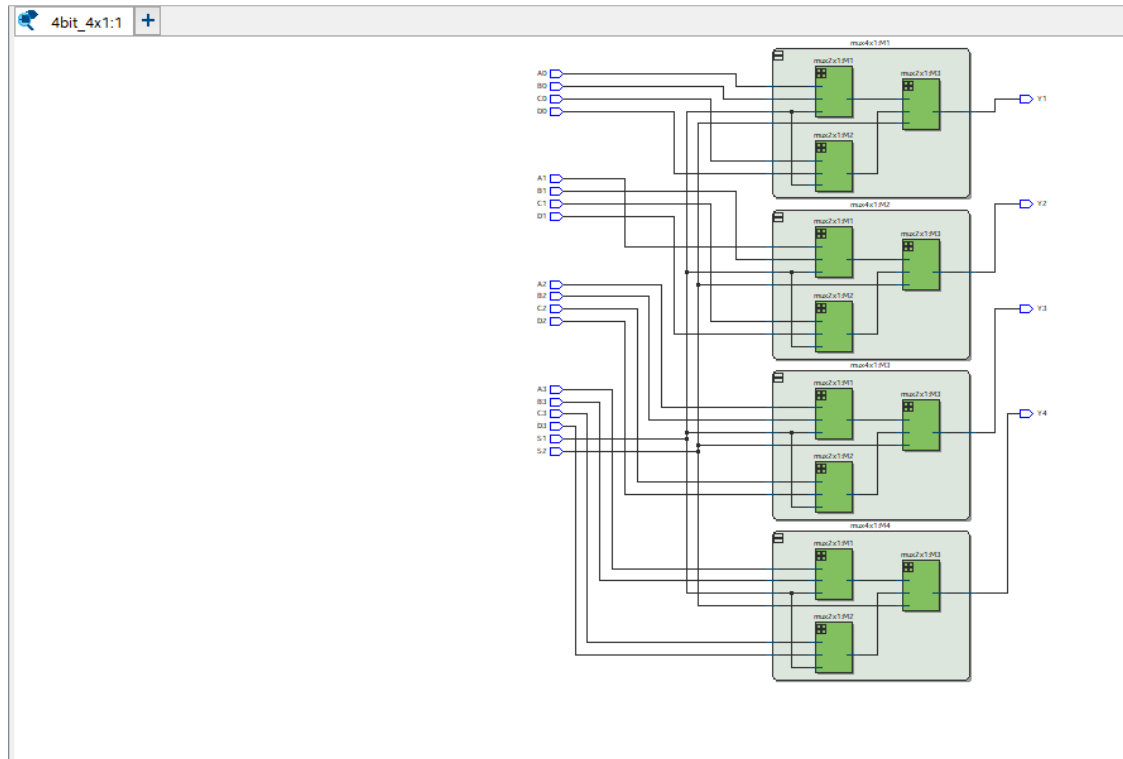


Figure 5: 4-bit 4x1 Multiplexer using 4x1 MUXs designed in Part B schematic

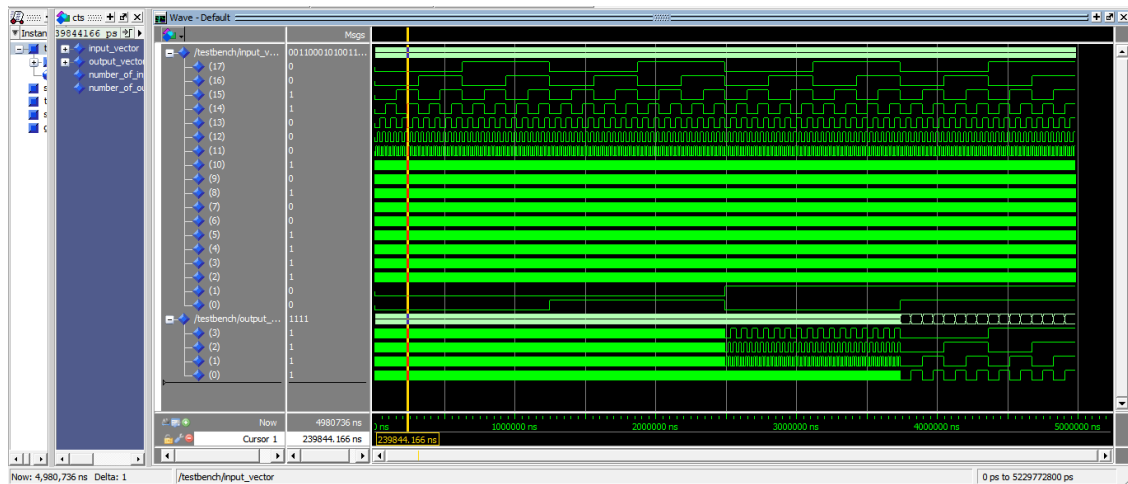


Figure 6: Corresponding waveform

```

Transcript
# run -all
# ** Note: SUCCESS, all tests passed.
#   Time: 4980736 ns  Iteration: 0  Instance: /testbench
VSIM 2>

```

Figure 7: Corresponding transcript